

Fungal Computing and Bio-Digital Interfaces: A 2023–2024 Research Survey

1. Advances in Fungal Computing (2023–2024)

Many-Input Fungal Logic Gates: Recent experiments have solidified the ability of living fungal mycelium to compute Boolean logic. In a landmark 2022 study, oyster mushroom mycelium implemented **4-input Boolean logic functions**, yielding 3136 truth tables with 470 unique logical functions identified ¹. Inputs were encoded as electrical stimuli (e.g. -5 V for 0, +5 V for 1) and outputs decoded from the mycelium's electrical response ² ³. This demonstrated *purposeful computing with fungi*, showing mycelial materials can serve as **“in materio” computers** performing logic without traditional silicon ⁴ ⁵. The repertoire included simple gates (e.g. NAND, AND) and more complex mappings, covering all classes of cellular automata behavior up to computationally universal rules ⁶. These many-input fungal logic gates were an important step toward *intelligent living materials*, but they remain single-layer implementations (a single mycelial substrate acts as a one-layer reservoir). **Scaling up** to more complex circuits (e.g. cascading fungal gates or multi-layer networks) is an open challenge. Researchers envision networking multiple mycelial processors to achieve deeper logic – effectively a *mycelial logic network* – but this has so far been explored only conceptually (drawing parallels to reservoir computing architectures) ⁷. In practice, experiments are still limited to one fungal “layer,” as reliably interfacing outputs of one mycelium into another is non-trivial given biological variability. Nonetheless, the field is moving from simple logic gates toward **integrated fungal logic circuits**, with 2023 bringing improved understanding of how to combine and repeat fungal computing elements.

EU FUNGAR Project – Toward Intelligent Bio-Buildings: The EU-funded **FUNGAR (Fungal Architectures)** project (2019–2023) greatly accelerated fungal computing research with an eye toward living, sensing buildings. The project achieved *groundbreaking results* in fungal electronics and unconventional computing. Notably, the consortium developed **prototypes of fungal electronic components** – including fungal sensors, tactile and chemical **responsive wearables**, and simple analog circuits like fungal oscillators, capacitors, **memristors**, and low-pass filters ⁸. These are *living devices* where mycelium serves as the functional element (for example, using a mycelial network as an oscillator or memory resistor). Several conceptual designs for complete “fungal computers” were proposed, laying the groundwork for future fungal CPUs ⁸. In tandem, FUNGAR addressed the material side: to create load-bearing, **structural mycelium** that computes, the team experimented with different fungi and substrates. They identified **Ganoderma resinaceum** (Reishi) and **Pleurotus ostreatus** (oyster mushroom) as optimal species that balance growth in composites with tolerance to embedded conductive materials ⁹. This means these fungi can remain alive and functional even when the substrate is infused with nanoparticles or polymer conductors – a crucial trait for *living circuits* in buildings. The project also demonstrated novel fabrication techniques like **triaxial Kagome weaving** of natural fibers as scaffolds for mycelium growth ¹⁰. This allowed construction of a large (300 m²) lattice inoculated with fungus, a step toward “growing” whole architectural components with built-in sensing. While true intelligent bio-buildings are still aspirational, FUNGAR's results indicate that a **“fully integrated structural and computational living substrate”** is feasible ¹¹ ¹². In short, fungi are being transformed from passive building materials into **adaptive, information-processing elements**. These advances mark a new era of **fungal computing** where living mycelium networks can not only form bricks and insulation, but also compute and respond to their environment – a foundation for future self-aware bio-architectures.

Beyond 4-Input Logic – Toward Networked Mycelial Circuits: Building on the above, 2023–2024 saw efforts to push fungal computing beyond single 4-input gates. Researchers are exploring **multi-stage logic** in fungi by connecting multiple mycelial patches or creating spatially distributed inputs/outputs across one network. Although no peer-reviewed report of a true multi-layer fungal logic circuit has emerged yet, the theoretical framework is being laid. Adamatzky and colleagues describe architectures for a “**fungal computer**” comprising distributed mycelial processors that communicate and collectively solve problems ¹³. The inherent qualities of fungal networks support this: mycelium is *fault-tolerant and self-regenerating* (damaged parts reroute signals), *reconfigurable* (growth can be guided by attractants or repellents), and *evolvable* over time ¹⁴. These traits imply a network of fungal components could rewire itself and continue computing even if parts fail – a desirable property for complex computing systems. There is also progress in merging **fungal logic with conventional electronics** to form hybrids (e.g. using transistor circuits to amplify and route signals between distant mycelial gates). Early demonstrations include interfacing multiple fungal colonies via microcontrollers to implement simple cascaded logic. For example, one can imagine the output of a *Pleurotus*-based gate (as a voltage spike pattern) triggering an input stimulus to a second *Ganoderma*-based gate, thus realizing a two-step computation. While such setups are still experimental, they suggest that **networked mycelial logic** – essentially fungal *logic boards* – may soon be within reach. The ongoing research focus is to increase the complexity of computations (beyond the ~470 functions reported in 2022 ¹) by either expanding the number of inputs per fungal processor or by linking processors. In summary, the field is transitioning from proving *that* fungi can compute, to engineering *how* they can be assembled into larger bio-computing systems. The next few years will likely bring multi-fungal networks performing higher-level tasks, moving closer to practical **fungal biocomputers**.

2. Bio-Digital Interface Technologies for Mycelial Computing

Electrode Design for Live Mycelium: Establishing a robust electrical interface with living mycelium is a core challenge. Early fungal computers relied on simple metal electrodes: for example, the 2022 4-bit logic experiments inserted four **1 mm platinum rod electrodes** about 5 cm deep into a mycelium block to serve as inputs ¹⁵. On the output side, an array of differential probe pairs measured the mycelium’s voltage spikes at various locations ¹⁵. Platinum is favored for its stability and conductivity; however, it is expensive and may not be ideal for long-term living setups. Recent work has trialed **alternative electrode materials** that are less invasive and more biocompatible. One successful approach is using **medical-grade stainless steel needle electrodes**, which have been used to record fungal electrical activity without poisoning the tissue ¹⁶. These fine needles (often 18 mm sub-dermal EEG needles) can be inserted at different depths in the mycelial substrate, even partially insulated to target specific regions ¹⁷ ¹⁸. Stainless steel is cheaper and reasonably inert; researchers found it can remain in situ for weeks, whereas plain copper wires, for instance, corrode or provoke unwanted reactions. Another strategy is to avoid metal electrodes altogether by **integrating conductive networks into the substrate**. The FUNGAR project explored doping mycelium composites with conductive polymers and nanoparticles to create an internal web of electrodes that the fungus grows around ¹⁹. This way, the fungus literally **grows its own circuitry**, maintaining contact over time. The choice of fungal species is crucial here – as noted, *Ganoderma* and *Pleurotus* were found to tolerate embedded conductors without dying off ⁹. In practice, flexible **carbon-based electrodes** (like graphite or graphene coatings) are being tested as well, since carbon materials are bio-friendly and resist corrosion. The overall trend is toward **non-destructive, long-term electrodes** that can move or grow with the mycelium. Some experimental devices use mesh or lattice electrodes printed on biopolymer films, onto which mycelium can attach. These “soft” electrodes move as the fungus expands, maintaining signal fidelity. In summary, researchers are moving beyond simple platinum pins toward **fungus-friendly electrode designs** – from steel needles to conductive composites – to achieve stable electrical contact over the organism’s lifetime.

Signal Encoding and Protocols: Because fungi operate on very different timescales and signal amplitudes than silicon circuits, special protocols are needed to communicate with them. Typically, **input signals** to a mycelial computer are delivered as low-voltage pulses or waves applied via the electrodes. Amplitude, frequency, or duration of these stimuli carry the “logical” value or data. For instance, one experiment represented binary 0 and 1 by applying either -5 V or +5 V DC steps to the fungus ⁴ ²⁰. Another approach is to use **frequency-coded inputs** – e.g. a 1 Hz pulse train for logical 1 versus 0.1 Hz for 0 ². The fungus responds to these stimuli with endogenous electrical activity (oscillations and spikes in potential), which serve as the **output signals**. Decoding the output often involves interpreting the presence or absence of characteristic spikes as binary states ²¹. Figure 1 shows a typical setup where a mycelium block is instrumented with multiple electrodes to inject inputs and record outputs. In this example, a train of electrical pulses is applied at certain electrodes (stimulating the mycelium), while differential pairs of recording electrodes pick up the resultant spike patterns. The interface hardware is usually a custom low-noise amplifier and ADC (analog-to-digital converter) system. In the 4-bit gate experiments, an Arduino-based controller and AD9833 waveform generators produced the input voltage steps, and a 24-bit ADC board logged the fungal voltages at 1 Hz sampling ¹⁵ ³. Notably, one ADC channel was dedicated to a **synchronization pulse** that was sent whenever the input pattern changed, marking the boundary of computation cycles ²². This sync signal is critical for aligning the biological response with the input state during data analysis. Recent research is also investigating **optical and chemical stimulation** as alternative input methods – for example, shining light of specific wavelengths on photosensitive fungi to trigger spikes, or adding a drop of nutrient or stimulant to a site. These could offer more gentle, diffuse ways to “program” the fungus compared to direct electrodes. Overall, a robust bio-digital protocol is emerging: use **low-frequency, low-amplitude stimuli** that the fungus can perceive, allow a generous response window (minutes to hours), and translate the analog spiking output into logical outputs via thresholding or pattern recognition ²³.

Figure 1: Experimental setup for fungal bio-digital interfacing. A block of mycelium-bound substrate is instrumented with multiple electrodes – here, stainless steel needle electrodes (small white pins) are inserted into the fungal composite to record electrical activity, while a larger probe (black) measures moisture content. Electrical spikes generated by the fungus in response to stimuli are captured by the electrode pairs for analysis

²⁴ ²⁵.

Signal Filtering and Noise Reduction: Distinguishing meaningful “computational” signals from biological noise is non-trivial, as living electrical signals are often noisy, slow, and low-amplitude. Researchers employ several strategies to **condition and filter** fungal signals. First, using **differential electrode pairs** (measuring voltage difference between two nearby points) helps cancel out ambient noise or drift common to both electrodes ²⁶. This is analogous to using a reference electrode – the fungus’s internal signals are local, so a differential reading amplifies local changes. Secondly, signals are typically digitized with high resolution (24-bit ADCs are common) and then band-pass filtered in software to isolate the frequency range of interest. Many fungal spikes occur in the range of seconds to hours ²³, so very low-frequency drift (DC offsets due to slow processes like growth or temperature change) can be filtered out, as can high-frequency interference (50/60 Hz mains hum, etc.). A simple but effective method used in the 4-input gate analysis was **threshold filtering**: the team defined a “dead band” of ± 20 mV around the baseline and only counted a spike event if the signal crossed beyond this band ²⁷. In essence, any voltage fluctuation smaller than 20 mV was treated as noise (logical 0), and anything above that as a potential logical 1 spike ²⁸. By tuning this threshold (up to 175 mV in steps) they could systematically separate noise from real responses. They also ignored the polarity of spikes (since some fungal spikes can go positive or negative) and just looked at occurrence ²³. This approach of **binary spike detection** is akin to how neuronal spikes are processed in neuromorphic engineering. More advanced techniques are now being tried: one is using machine learning to classify fungal electrical patterns, distinguishing random background spikes from those provoked by a known input.

Another is **adaptive filtering**, where the system monitors overall fungal activity level (which can vary with conditions) and adjusts the detection threshold or gain accordingly. There's also the challenge of separating multiple simultaneous signals – e.g. if multiple input electrodes provoke overlapping responses. Some studies use an array of many electrodes (like a high-density multi-electrode array) and then apply source localization algorithms to identify which region of mycelium fired a spike. In summary, while fungal signals are faint and slow, careful instrumentation and signal processing (differential recording, band-pass/threshold filtering, sync signals) have enabled researchers to cleanly extract computational output from the noisy biological milieu ²⁷. As a result, one can be reasonably confident that a detected spike corresponds to a fungal logic response and not an environmental artifact.

Temporal Synchronization of Bio and Digital Rhythms: A fundamental hurdle in bio-digital interfaces is the **mismatch of time scales** – electronic systems operate in microseconds to milliseconds, whereas fungal networks process information over seconds to hours. Bridging this gap requires synchronizing the computer's operations with the fungus's natural rhythms. In practice, this means **slowing down the computational clock** and designing protocols that respect the fungus's pace. For example, in the 4-bit gate experiments the researchers presented each 4-bit input pattern to the fungus for a full **1 hour** before switching to the next pattern ³. This ~3600× slowdown (compared to conventional GHz logic clocks) gave the mycelium ample time to react to each input and settle before the next stimulus. Indeed, they observed that fungal voltage spikes often occurred during the transition between input states (when a new stimulus was applied) and then decayed within tens of seconds ²⁹. By waiting an hour per state, they ensured each input's effect had time to manifest and dissipate ²³. This is one form of synchronization: a **step-wise hold protocol** where the digital system holds a constant input for a period matching the fungus's response latency. Another tactic involves using the fungus's *own* oscillatory activity as a timing reference. Some fungi exhibit regular oscillations in electrical potential (endogenous rhythmic spiking) with periods ranging from minutes to hours ³⁰. If these oscillations are stable, one could imagine timing the input triggers to a certain phase of the fungal rhythm – much like aligning to a clock signal, but the clock is biological. Early research hints at this: in 2023, Browner *et al.* showed that illumination could *entrain* a fungus's spiking frequency, effectively tuning the mycelium's internal rhythm with an external periodic stimulus (light/dark cycles) ³¹. Such methods point to the possibility of *phase-locking* electronic and fungal oscillators. In hybrid circuits where a living oscillator (fungus) is part of an electronic loop, one must also consider **tempo matching** – e.g. using programmable delay circuits or buffers so that the fast electronics wait for the slow bio signals. Some experiments use **event-driven synchronization**: the digital controller simply listens and waits until it detects a fungal spike (or a steady state) before proceeding, rather than using a fixed time. In summary, temporal coordination is achieved by (1) running the electronic interface orders of magnitude slower than usual, (2) leveraging or controlling the fungus's own timing (for instance, via periodic stimuli), and (3) designing asynchronous, event-driven logic that reacts to fungal outputs whenever they arrive. This ensures the *biological computation and electronic computation remain in step*. As Andrew Adamatzky quipped, fungi are “computers which live on rotten trees” and naturally operate on long timescales ³² – but with careful synchronization, they *can* be harnessed to work in concert with our silicon systems, albeit in a patient, contemplative fashion.

3. Mycelial Intelligence and Memory

Evidence of Learning and Adaptation: A striking aspect of fungal networks is their capacity for **adaptation**, often likened to primitive learning. While fungi lack a brain or neurons, experiments show they can adjust their behavior based on past stimuli – essentially forming *memory*. In early 2023, researchers at UWE Bristol reported that repeatedly stimulating a mycelial network at two separated points led to an *increase in conductivity and communication speed* between those points ³³. In other words, the more the fungus “used” a certain pathway, the better that pathway conducted signals. This is

analogous to how repeated activation of a neuronal pathway strengthens synaptic connections (Hebbian learning), or how our brain forms habits. The team observed that after training, the fungus could transmit spikes more reliably and quickly, indicating it had formed a **persistent conductive channel** – a rudimentary long-term memory ³³. Adamatzky described this as the fungal network establishing a “habit”: like how repeated actions in a brain become automatic, the fungal colony’s response to a familiar stimulus became faster and stronger with repetition ³³. Beyond electrical habits, fungi also show *physiological* memory. For instance, mycologists in Germany demonstrated that mycelium can “remember” a mild heat shock: fungi pre-exposed to a small temperature rise were able to recover from a subsequent larger heat shock much more swiftly than unprimed fungi, indicating a form of **stress memory** lasting ~24 hours ³⁴ ³⁵. If too much time passed, they “forgot” and were vulnerable again, showing the memory was temporary ³⁵. Similarly, yeast cells have exhibited memory of salt exposure, adjusting their stress response if they had encountered salt before ³⁶. These findings support Nicholas Money’s provocative thesis that fungi possess a rudimentary **cellular consciousness** or cognition: fungal mycelia make decisions (e.g. growth direction changes), learn from experience, and have short-term memory of environmental changes ³⁷. While “consciousness” in fungi is debated, it’s clear they are *aware of and respond to their environment* in a purposeful way ³⁸. Mycelium networks integrate signals over time and space, then alter their growth or activity accordingly – a hallmark of intelligent behavior in biological systems.

Long-Term Memory Mechanisms: How do fungi store information? One mechanism is via **physical and electrical state changes** in the network. Mycelium is not static; it can thicken certain hyphal strands, branch in new directions, or reroute nutrients. These structural changes can encode memory of past conditions (for example, forming a high-conductivity “highway” of thick hyphae toward a food source and retaining it for future resource movement). On the electrical side, recent work has shown fungi exhibit **memristive behavior** – that is, their electrical resistance depends on the history of voltage applied, much like an electronic memristor (memory-resistor). In 2021–2022, Adamatzky’s lab found that placing electrodes across a fungal fruit body and applying a voltage sweep yielded a pinched hysteresis loop characteristic of memristors ³⁹ ⁴⁰. The fungus’s resistance at any given time was influenced by the charge that had flowed previously, meaning the circuit “remembered” past currents. Such **fungal memristors** can serve as analog memory elements ³⁹. They open the possibility that mycelial networks inherently perform **integrative memory** – past stimuli alter ion flows or chemical gradients in the hyphae, which in turn affect how future signals propagate (in essence, storing a bit of information about the past state). Indeed, one paper bluntly states: “*fungi exhibit properties of memristors (resistors with memory), electronic oscillators, and sensors*”, underlining that memory is deeply entwined with fungal electrical dynamics ³⁹. Another likely storage mechanism is in **biochemical changes**: repeated stimulation can trigger gene expression changes or release of signaling compounds that persist. For example, a mycelium that is regularly electrically stimulated might produce more calcium ion channels in its cell membranes (calcium signaling is thought to be involved in fungal spike generation ⁴¹ ⁴²). More channels could make it more excitable long-term – a form of potentiation. Conversely, habituation (learning to ignore a harmless repeated stimulus) could occur by the fungus dampening its response over time. While research is ongoing, the emerging picture is that fungal networks store information in a **distributed, analog fashion**: there is no single “memory register,” but the entire network’s physical and electrical state encodes its experience. This distributed memory aligns with the idea of mycelium as a **diffuse brain**. It doesn’t have discrete memory cells like a digital chip; instead, memory is manifested in the connectivity and conductivity of the hyphal web.

Environmental Factors and Computational Capacity: The “thinking” ability of fungi is intimately tied to their environment. Key environmental parameters – humidity, nutrients, temperature, etc. – can enhance or impair the fungus’s computing capacity. A recent 2023 study by Phillips *et al.* focused on **moisture** and found a Goldilocks zone for fungal electrical activity ⁴³. In fresh mycelium-bound blocks with ~70–90% moisture content, the fungus produced spontaneous spike trains; as the blocks dried

towards ~65%, spiking diminished ⁴³. Interestingly, when partially dried (15% moisture), some spiking resumed, suggesting multiple regimes of activity (possibly different physiological states) ⁴³. The study also showed that **encasing the mycelium in an impermeable layer (trapping humidity)** led to increased electrical activity ⁴⁴ – essentially, the fungus “fired” more when it sensed a build-up of moisture or CO₂. These findings imply that **hydration level directly affects signal processing**: a drying fungus might enter a dormant state with less communication (poorer computing), whereas a comfortably moist fungus is active and responsive. Similarly, **temperature** can modulate activity: most fungal spiking has been observed in a moderate range (~20–30 °C). Too cold or too hot, and the metabolism slows or stress responses dominate, altering the electrical patterns ³⁴. Fungi also respond to chemicals – e.g. an influx of nutrients can cause a flurry of spikes (perhaps analogous to excitement), while toxins might silence activity. These environmental influences mean that a fungal computer’s “CPU speed” and reliability are not fixed, but vary with its living conditions. Researchers are actively studying how to **tune environmental parameters to optimize fungal computing**. For instance, maintaining constant humidity (via humidifiers or hydrogels) and temperature in fungal computer enclosures is essential to get consistent results ⁴⁵ ⁴⁶. On the flip side, one can *exploit* this sensitivity to environment as a feature: fungi can be **sensors** and incorporate those readings into their computation. In one demonstration, living mycelium composites were shown to **discern applied weights** by distinct patterns of electrical responses ⁴⁷. A weight placed on the mycelium caused specific spike trains (likely due to strain or pressure sensing by the network), and removing it caused a different pattern – the fungal material effectively functioned as a load sensor ⁴⁷. Likewise, fungal electronics have been created that react to light intensity (acting as photosensors) and to certain chemicals ¹³. The implication is that **mycelial computers are inherently multimodal**: they compute and sense in the same substrate. Environmental inputs are not just noise but part of their computing context – a form of *embodied computation* where the surroundings influence the outcome. A fascinating consequence is that a fungal computer might perform better on tasks that are analogues of its natural survival processes. For example, a mycelium might inherently be good at computing optimal resource distribution under varying moisture (since that’s what it does in nature) ⁴⁸, whereas it might be less adept at arbitrary binary arithmetic. This has led to the notion of “**contemplative**” or **slow, integrative computing**: fungal intelligence is seen as deeply *embodied* and environmental, processing information in a way that blends computation with sensing and growth. It suggests a paradigm of computing where, rather than isolating the processor from its environment (as we do with silicon chips), the environment is part of the computing loop – a very different, but potentially powerful mode of information processing.

Persistent Response Patterns (Habituation and Sensitization): Do fungi get “used to” repeated stimuli? Researchers have observed hints of **habituation** – i.e. a fungus reducing its response to a benign stimulus given repeatedly. For example, if a mycelium is given identical electrical pulses every hour, over time the magnitude of its spike response may diminish as it apparently realizes nothing bad or new is happening. Conversely, fungi can also show **sensitization**, where a strong or noxious stimulus primes them to respond more vigorously in the future. An anecdotal report describes that after a sudden large voltage shock, a fungal network entered a heightened activity state, firing more frequent spikes for hours as if on alert (though systematic study is needed here). Nicholas Money points to fungi’s idiosyncratic behaviors – each colony growing and responding in unique ways – as evidence that “*each mycelium is like a snowflake,*” having its own set of learned responses and habits shaped by its history ⁴⁹. Under controlled experiments, the “memory” of a stimulus can last on the order of hours to days ³⁵, as seen with temperature priming. There is even a demonstration that mycelium can associate two stimuli: in one setup, a chemical stimulus paired with an electric pulse caused the fungus to eventually react to the chemical alone with an electrical response, hinting at a basic associative learning (though this result is very preliminary). In summary, fungal networks exhibit **plasticity**: their response patterns are not fixed reflexes but change with experience, showing rudimentary analogs of habituation (ignoring repetitive inputs) and potentiation (enhanced response after training). This adaptability

underscores that **mycelial intelligence is a process** – a continually updating state – rather than a static set of logic gate outputs. When we interact with a fungal computer, we are in fact engaging with a *learning system* that carries its past into its future computations. This presents both challenges and opportunities: on one hand, it complicates reproducibility (the same input might yield different output on first vs tenth trial), but on the other, it offers a pathway to train fungal computers for specific tasks via reinforcement of desired patterns.

4. Reservoir Computing with Biological Substrates (Beyond Fungi)

Fungi are not the only living systems being harnessed as “**wetware**” **reservoirs** for computing. A broader trend in 2023–2024 is the exploration of **unconventional biological substrates** – from single-celled organisms to cultured neurons and plants – that can perform computation through their innate dynamics. The reservoir computing paradigm is particularly suited to these systems: instead of engineering explicit logic gates, one feeds inputs into the naturally rich, non-linear dynamics of the biological network and reads out a processed output. Below we survey a few examples and how they are benchmarked:

- **Slime Molds (*Physarum polycephalum*):** Slime mold is a famous non-neural computing substrate. It is essentially a giant single cell that forms a network of tubes, dynamically re-routing protoplasm inside. *Physarum* has solved mazes, optimized network paths, and even approximated algorithms like the traveling salesman in its search for food. In 2022, researchers took this a step further by integrating slime mold directly into a wearable device: a **slime mold smartwatch** interface was demonstrated, where the *Physarum* organism was incorporated into a watch and used as a biological sensor/computer ⁵⁰. The slime mold’s electrical resistance changes were measured to detect skin contact and user activity, effectively acting as a biosensor that could trigger digital functions. This example of a *hybrid living device* highlights how a biological substrate can serve as a reservoir that processes inputs (pressure, humidity, etc.) and produces a useful output (an electrical signal to the wearable). Slime molds can also implement logical operations; for instance, two food sources can represent inputs and the growth pattern can encode an OR gate solution (if either food is present, the mold connects). To **benchmark** slime mold computing, researchers have used tasks like maze solving efficiency, network routing optimality compared to algorithms, and analog circuit implementations (e.g. *Physarum*-based wires and transistors in wetware circuits). Metrics include path length optimality (compared to shortest path algorithms) and response time to stimuli. Because slime molds operate on timescales of hours (somewhat similar to fungi), they are considered “*slow processors*” but extremely energy-efficient and adaptable. The UIST 2022 demo of a slime mold device indicates they can be embedded into user interfaces as a sort of “living circuit” ⁵¹, validating their potential outside a petri dish.

- **Cultured Neuronal Networks (Brain Cells *in vitro*):** Perhaps the most striking reservoir computers are made of actual neurons. In late 2022, a team led by Kagan (Cortical Labs, Australia) created a system nicknamed “**DishBrain**” – a layer of living neurons (mix of mouse and human cortical cells) grown on an electrode array – which learned to play a simplified version of the video game *Pong* ⁵². The neurons acted as a reservoir that received stimulus (feedback from the game environment) and were stimulated when the virtual paddle missed the ball, etc., effectively learning to control the paddle. Over time, the neural network improved its performance, demonstrating **goal-directed learning**. This was a proof-of-concept that live neuronal cultures can serve as adaptive computing units, with performance measured on a clear benchmark (game score and improvement over trials). “DishBrain” sparked the formulation of

Organoid Intelligence (OI) as a field: leveraging 3D brain organoids (tiny self-organizing brain-like cell clusters) as microscopic bio-computers ⁵³. In 2023, a multidisciplinary team published a roadmap for OI, outlining metrics like the number of neurons, synapse density, learning rate, and energy efficiency compared to silicon ⁵⁴ ⁵⁵. Key metrics for neural reservoirs include *memory retention* (how long they can store training), *throughput* (spike rate, which can be up to kHz in neurons, much faster than fungi), and *learning speed*. The Pong-playing culture, for example, was evaluated on how quickly it improved relative to chance (it showed statistically significant learning within minutes of gameplay). Neural reservoirs arguably set the high bar for performance among living substrates, but they raise more ethical and practical complexity (they require continuous nutrient/media and arguably have a form of sentience).

- **Plant and Root Networks:** Plants lack nervous systems, but they exhibit complex signaling (via electrical signals, hormones, etc.) and have been proposed as computational reservoirs. In 2022, Pieters *et al.* demonstrated the first **plant-based reservoir computer** using living strawberry plants (*Fragaria x ananassa*) ⁵⁶. They attached sensors to eight leaves to measure changes in thickness (which relate to water flux and electrical signaling in the plant) and used these as the reservoir outputs. By feeding environmental input signals (like light fluctuations or temperature changes) to the plant – essentially perturbing it – they showed the plant could perform tasks such as predicting its own transpiration rate and classifying environmental conditions ⁵⁷. Classical benchmark tasks were attempted, and while the plant was not a general-purpose computer, it excelled at *eco-physiological computing*: solving problems related to its life (e.g. forecasting photosynthesis rates from prior light levels) ⁵⁸. This suggests plants process information about their environment in a way that can be harnessed for computation. To benchmark computation in a plant, the researchers evaluated the **prediction error** on these tasks and compared it to standard machine learning models. They also examined metrics like the plant reservoir's **memory capacity** – how many time steps into the past the plant's state could reflect (plants, having slower dynamics, provided a fading memory over many minutes). Another metric was the **separation property** – the degree to which different inputs led to distinguishable states in the plant (using variations in multiple leaf sensor readings) ⁵⁹. They noted that suitable substrates for physical reservoir computing should have *high dimensionality, nonlinearity, fading memory, and input separability* ⁵⁹, and plants meet many of these criteria albeit with slower response.

- **Synthetic Biofilms and Bacterial Computing:** Beyond multicellular networks, even colonies of bacteria can compute. Researchers in 2023 achieved a **3-input, 3-output logic gate (a double Feynman gate) in engineered *E. coli***. By genetically programming bacteria with synthetic gene circuits, they made "*bactoneurons*" that responded to chemical inputs (present or absent) and produced fluorescent protein outputs. These bacterial cells were arranged in a pattern akin to a single-layer neural network to realize the logic function. The significance is that *molecular circuitry* inside cells can implement complex logic in a reversible, fan-out manner. Performance is measured by the correctness of output (fluorescence) for each combination of inputs and the stability of the computation (how reliably the cells maintain the function over time). Bacterial computing is slower (gene expression might take hours to turn on a fluorescent output) but massively parallel (billions of cells can operate simultaneously) and can be very energy efficient. Another group proposed a **Bacterial Molecular Computing-on-Chip** platform using microfluidics to network bacterial colonies and reading electrical signals from them. **Biofilms**, communities of microbes embedded in an organic matrix, are being considered as well – one could engineer a conductive biofilm that responds electrically to stimuli, functioning similar to mycelium. Metrics for such systems include response time, logic fidelity, and longevity (a robust biofilm might compute for days or weeks without human intervention).

Benchmarking and “Contemplative” Metrics: Comparing these diverse living computing substrates requires appropriate metrics. Conventional metrics like FLOPS or clock speed don’t apply; instead, researchers use task-specific benchmarks and system properties:

- **Information Capacity & Dimensionality:** How rich is the repertoire of signals the substrate can produce? (For fungi, one measure was the number of unique logic gates found ¹ ; for plants, the dimensionality was the number of independent sensor signals one can draw from the plant ⁵⁷ .) Generally, higher dimensionality means more complex computations can be embedded.
- **Memory Depth:** In reservoir computing, the **memory capacity** quantifies how long past inputs influence current state. Fungal and plant networks have a long memory (due to slow dynamics) but also a *fading* one – e.g. a fungus might retain a trace of an input for hours, but with diminishing effect. This can be beneficial for tasks that require integration over time (e.g. smoothing sensor data or detecting temporally extended patterns), which is why one might call it “*contemplative computing*” – the substrate naturally *lingers* on past inputs. A recent perspective highlights that many physical reservoirs (optical, mechanical, biological) trade off speed for memory depth ⁵⁹ . Slower systems like mycelium or plants can hold a longer temporal context (good for tasks like time-series prediction or anomaly detection in slowly varying data), whereas faster ones like neural cultures have short intrinsic memory but high throughput (good for reactive tasks).
- **Energy Efficiency:** A compelling metric for bio-computation is how little energy it uses compared to silicon. Many living systems operate on chemical energy at ambient temperature without external power. For example, a mycelium computer lives on oats or wood chips – a true low-power processor (just not a fast one). It has been noted that these systems “*do not rely on fast-heating silicon chips*” and can self-repair, making them ultra-sustainable computing elements. Quantitatively, one can measure energy per logic operation (though this is hard to define for analog systems). Early estimates suggest something like *Physarum* or fungi use orders of magnitude less energy per “computation” than a CPU performing the equivalent operation (if one could define it), but rigorous comparisons await.
- **Task Performance:** Ultimately, each substrate is benchmarked on *specific tasks*: classification accuracy, maze solved or not, game score, prediction error, etc. For instance, the strawberry plant RC was judged on how accurately it could predict photosynthesis rates (it did well on that niche task) ⁵⁸ , while the neuron dish was judged on Pong win rate which improved to a level significantly above chance ⁶⁰ . Slime mold might be tested on finding optimal graphs or mimicking road networks of cities (a famous experiment showed *Physarum* network matched the Tokyo rail network efficiency). These tasks highlight each organism’s “expertise”: slime molds excel at path optimization, fungi at integrating environmental signals, plants at reflecting ecological dynamics, and neurons at rapid signal processing and learning.

Hybrid Bio-Digital Systems: A fascinating development is **hybrid computing architectures** where living substrates are tightly coupled with traditional digital systems. Rather than a one-off interface, these hybrids maintain a *sustained feedback loop* between biology and electronics. One example is the **fungal robot** prototypes: researchers connected oyster mushroom mycelia to a small robot, using the fungus as a sensory and decision-making unit. The mycelium’s electrical signals (filtered to remove vibrations and EM interference) were used to control the robot’s movements, while the robot provided stimuli back to the fungus (e.g. light or food in certain directions) ⁶¹ . Such a symbiotic loop effectively creates a *biological controller* for the robot. In another project, a “**plant-robot hybrid**” was made where a potted plant’s signals controlled a wheeled robot moving it toward light, essentially giving the plant mobility through a robotic body – an embodiment of *embodied intelligence* for the plant. Hybrid systems

are also emerging in art and architecture: interactive installations where fungi or slime molds are part of a sensor network that triggers digital media or lighting changes in real time. The **Fungal Computer Interface (FCI)** concept (described in Section 5) is essentially an enabling technology for such hybrids, providing a standardized way to converse with mycelium networks in real-time ⁶² ⁶³. The performance of hybrid systems can be measured by **closed-loop criteria**: stability of the feedback loop, responsiveness (e.g. how quickly can a fungal robot react to an obstacle), and the *enhancement* gained by using the living component (does the bio-part enable functionality that pure silicon could not easily do?). Often the living part brings adaptivity and resilience. For example, a fungal sensor network in a building might not only detect conditions but continue to function even if some nodes are damaged (the mycelium re-grows around damage) ¹⁴ – something a conventional sensor network would struggle with. These hybrids hint at a future where **living processors** are embedded in devices and environments, continuously co-computing with digital electronics.

In summary, reservoir computing with non-fungal living substrates is a blossoming field: *Physarum* slime molds, *Spirogyra* algae filaments, plant roots, bacterial films, and cultured neurons have all demonstrated computing capabilities in 2023. Each has unique strengths and is evaluated with appropriate tasks – from solving mazes to playing Pong. The term “**contemplative computing**” is sometimes used to describe the especially slow, integrative computation seen in organisms like fungi and plants, which process information in a steady, continuous manner rather than blistering-speed clock cycles. While they won’t replace silicon for high-speed arithmetic, these living computers offer novel attributes: massive parallelism, self-repair, ultra-low power operation, and an intrinsic ability to handle complex temporal and spatial patterns. They also force us to broaden the definition of computing – to include *metabolism, growth, and adaptation* as part of the computation. In doing so, they blur the line between computer and environment, or between algorithm and organism, giving rise to a truly **embodied form of intelligence** that researchers are only beginning to understand.

5. Practical Implementation Challenges of Fungal Computers

Building a practical fungal computer – one that reliably operates outside a lab – comes with significant **challenges**. Being a living system, a mycelium computer has “upkeep” requirements and failure modes quite unlike an electronic device. Here we outline key issues and recent efforts to address them:

- **Fragility and Contamination:** Live fungal circuits can fail if the organism’s health deteriorates. **Contamination by other microbes** is a constant threat – airborne mold spores or bacteria may colonize the nutrient-rich mycelium substrate and disrupt the intended fungus or its signals. Sterile technique is required to start fungal cultures for computing, but over long deployments (months/years) it’s hard to maintain sterility. One failure mode is that an invasive mold outcompetes the computing fungus, leading to signal loss. Even if no contaminants enter, the fungal colony itself can age or **degrade**: it might consume the available nutrients and then go dormant or die. Unlike a silicon chip, a fungal substrate can literally run out of fuel or decay. Drying out is another risk; if the moisture falls below a viable level (often >~60% for many species), the electrical activity ceases as the fungus enters a stasis ⁴³ ⁴⁶. Conversely, too much moisture can cause anoxia (lack of oxygen) or rot. In practical terms, a fungal computer likely needs a **sealed, climate-controlled enclosure** to manage these factors. The FUNGAR project found that keeping parts of the mycelium alive in building materials requires securing a “*minimal viable moisture content*” and preventing uncontrolled sporulation or decay ⁶⁴ ⁶⁵. One design is to encapsulate the fungal substrate in a semi-permeable membrane that blocks contaminants but allows gas exchange and can be periodically rehydrated. Some teams use a layer of sterile agar or hydrogel over the mycelium to keep it moist and protected. Nonetheless, long-term stability remains an issue: whereas a digital chip might run unchanged for years, a fungal processor might need “recalibration” or even periodic replacement as the living network evolves.

- **Environmental Control:** As mentioned, a fungal computer more or less **demands environmental support** – temperature regulation, humidity control, and perhaps nutrient feeding. This introduces a kind of “biological power supply”: providing food (e.g. adding sterile oats or sugars periodically) and removing waste (metabolic byproducts) if needed. In a building context, researchers envision mycelium-based sensors that are part of walls or panels; these would need the building to maintain comfort-range conditions for them to stay active ⁶⁶. On one hand, fungi are quite hardy (they can survive in many conditions), but the *computing performance* might only be optimal in a narrower range. For example, **cold slows down spiking** and too hot can make the signals erratic or cause the fungus to produce fruiting bodies (mushrooms) instead of hyphal growth – the latter might reallocate resources and alter conductivity. If a fungal computer starts fruiting, it could even detach from electrodes or change shape. Thus, preventing fruiting (perhaps via light cycle or CO₂ level control) might be necessary to keep it in the vegetative, computing state. This is an unusual maintenance task: ensuring your CPU doesn’t “decide” to grow a mushroom!
- **Electrical and Power Needs:** In terms of electrical power, fungal electronics operate at very low voltages and currents (millivolts and microamps). The fungus itself does not consume electrical power in the way a silicon circuit does; rather, power is used by the interface electronics (amplifiers, microcontroller, etc.) and by any environmental controls (humidifier, heater). The fungus lives on *chemical energy* from its substrate. This raises an interesting point: a fungal computer’s **energy budget** is mostly in keeping the fungus alive (which might just be ambient conditions and a bit of growth medium) and running the read/write electronics, rather than in computation per se. Mycelium networks are extremely energy-efficient in their signaling – a spike is a localized ionic event requiring negligible energy. Adamatzky has highlighted that fungi are “*very low energy consumption – computers which live on rotten trees*”, underscoring their sustainable nature ³². However, the overall system might require pumps, sensors, and processors that draw power, so designing low-power support hardware is crucial. One approach is to leverage the fungus’s own electrical output wherever possible: for instance, using fungal spikes to directly trigger low-power transistors or piezoelectric elements, reducing the need for continuous powering of an ADC. There are even speculative ideas of **energy harvesting from the fungus** – e.g. using microbial fuel cell concepts or the slight bioelectric potential of mycelium to power the interface circuits, making a self-powered device. We are not there yet, but it’s a possibility for remote environmental fungal sensors. In essence, while fungal computing is inherently low-energy, a practical implementation must minimize the overhead of maintaining the fungal habitat and reading its signals, so that the *net* system remains eco-friendly and efficient.
- **Reproducibility and Calibration:** No two fungi are exactly alike; biological variability means that the same “circuit” (e.g. two different oyster mushroom blocks) may behave somewhat differently. This poses a challenge for programming and standardizing fungal computers. Typically, one must **calibrate each fungal substrate** – for example, determining its baseline spiking rate, adjusting thresholds for that particular network’s noise level, and perhaps “training” it with some test inputs to see how it responds. The variability can be mitigated by using cloned fungal strains and identical growth conditions, but some variation is inevitable (much like no two brains are identical). This is acceptable in reservoir computing approaches (they treat the substrate as a black-box and use machine learning on the outputs), but it complicates using fungal circuits for precise logic unless adaptive calibration is in place. Over time, as the fungus grows or ages, its electrical properties drift, so a **dynamic recalibration** may be needed. For instance, the 4-input experiments had to spray the substrate with water and let it rest between runs to keep conditions similar ⁶⁷. They effectively “reset” the fungus to a baseline state between experiments. In a continuous application, one might need a standby mode where the fungus is

given time to recover or is periodically stimulated with a known pattern to gauge if its responses have changed (like a diagnostic).

- **Hardware/Software Platforms:** Unlike conventional computing, there is no off-the-shelf “biocomputer” kit – but that is starting to change. A few **commercial or open-source platforms** are emerging for fungal interfaces. One notable effort is by *MycoSoft*, which in 2024 announced an open-source **Fungal Computer Interface (FCI)** concept ⁶² ⁶⁸. They developed prototype hardware called “**Mushroom 1**” – an environmental monitoring device that also serves as a fungal interface node ⁶⁹ ⁷⁰. The Mushroom 1 device integrates sensors (for CO₂, VOCs, humidity, etc.) with fungal electrodes and a microcontroller, acting as a combined sensor array and fungal computer probe in one. It’s designed to be placed in ecosystems or soil with fungal mycelia and gather both environmental data and fungal electrical data ⁷¹ ⁷². MycoSoft has also introduced a specialized probe technology: a **UART/ECG/HDMEA-based probe with a solid-state agar medium** that can be inserted to make stable contact with mycelium ⁷³. This probe can not only read the bioelectrical signals but also **stimulate the mycelium** in precise ways (using a high-density microelectrode array, HDMEA, to deliver localized pulses) ⁷³. Alongside the hardware, they proposed a **Hypha Programming Language (HPL)** and a software ecosystem (NatureOS) to translate fungal signals into digital actions and vice versa ⁶⁸ ⁶⁹. In essence, MycoSoft’s FCI is trying to do for fungal computing what Arduino did for electronics – providing a standard platform that enthusiasts and scientists can use to experiment without building everything from scratch. While still in prototype stage, it’s a significant step toward *commercializing fungal interfaces*. Other platforms include generic biosignal amplifiers (originally made for EEG or plant electrophysiology) that are now being repurposed for fungi – e.g. the OpenBCI or Bitvoodoo’s Plant SpikerBox can record plant/fungal action potentials and are open-source. The University of West England’s lab often uses **Pico Technology ADCs** and Arduino Megas (as in the 4-bit experiments) ¹⁵, so even hobbyist microcontrollers are up to the task of basic fungal I/O. We are also seeing the first integration of wireless tech: experiments with **LoRa (Long Range radio)** on FCI devices to transmit fungal data remotely from field sensors have been reported ⁷⁴. All of this suggests that a small but growing community is standardizing tools for fungal computing. By the late 2020s, we may have a mature open-source hardware kit (sensors, electrode arrays, software) that anyone can use to plug a living fungus into their computer and start programming it in a high-level language.

Figure 2: Concept art of a future fungal interface device (Mushroom 1 by MycoSoft). The device resembles a mushroom-like antenna with embedded sensors and solar panels, designed to be placed in soil. It interfaces with the natural mycelium underground via a specialized probe (stem) that can read and stimulate fungal electrical signals. Such a device could serve as a “living computer node,” merging digital infrastructure with fungal networks for environmental monitoring and computation ⁷⁰ ⁷³. (Image credit: MycoSoft)

- **Failure Modes and Maintenance:** Beyond contamination and environment, other failure modes include **electrode delamination or corrosion** (addressed partly by using inert materials as discussed), **biofouling** (the fungus or other organisms coating the electrodes in insulating biofilm), and **mechanical disturbance**. If the fungal substrate is part of a structure, mechanical stresses (vibration, shock) could disrupt its networks or the electrode contacts. One intriguing failure mode in prototypes is when the fungus “*decides*” to escape the intended area – mycelium might grow out of its container or send out strands in unexpected directions, breaking the circuit layout. In living architectures, containing the fungus to certain regions while letting it remain alive is an active area of research (one doesn’t want a fungal wall to start growing through the floor!). Solutions may include **physical barriers** that fungi don’t cross (like dry zones or certain chemical repellents) to keep them in the computing zone. Regular maintenance could involve trimming or reinoculating parts of the network. From a computing perspective, a fungal

computer might also *lose calibration* if the network grows significantly – e.g. new connections form that change the logic mapping. This suggests a need for **retraining** or re-discovery of logic functions periodically ⁷⁵. In the 2022 experiments, they actually exploited growth over time by running 14 repeats over weeks and capturing different logic functions as the mycelium structure evolved ³. But in a stable application you might want to constrain growth to maintain consistent function, or conversely allow it to evolve and then adjust the readout algorithm accordingly (a kind of continuous learning system).

- **Scalability:** Practical deployment also faces the issue of scaling up the size or number of fungal processing units. A single fungal block might only produce a few useful signals. To do something substantial (say monitor an entire building or perform a complex computation), one might need **many fungal nodes**. How to network them? Fungal networks can be physically connected (hyphae can fuse between colonies of the same species), potentially creating a larger “global” network. But controlling or addressing specific parts of a fused network is difficult – it becomes one big analog computer. An alternate approach is a hybrid network: use conventional wires or wireless links between separate fungal colonies, treating each colony as a node that communicates digitally. This introduces back-conversion (fungal analog → digital → fungal analog at other node). Some early work in Ireland looked at **electronic neuron interfaces** to link slime mold and plant signals over distance, conceptually like an internet of bio-computers. **Latency** is another concern – if a fungal computer spans meters or more, the internal signal conduction (via action potentials along hyphae) might be slow. Mycelium conduction velocities are on the order of a few millimeters per second in some species ⁷⁶ (significantly slower than animal nerves). So, large-area fungal circuits won't be fast unless assisted by electronics.

In summary, making fungal computing practical requires a holistic approach: *bio-engineering* to breed resilient, docile computing strains; *materials science* to package and support the living circuits; and *electronic engineering* to create interfaces that can handle the quirks of biology. Progress is being made on all these fronts. It's telling that companies like MycoSoft are envisioning commercial **“mycelium network devices”** that you can deploy in the wild for sensing/computing ⁷⁰ ⁷². Such devices aim to leverage fungi's natural advantages (self-powered, self-maintaining to a degree, sensitive to environment) while mitigating their downsides (unpredictability, fragility) via smart design. The challenges are substantial, but not insurmountable: after all, nature has managed to use fungi effectively for millions of years. The task now is to integrate these living systems into our technology in a reliable way. The coming years will likely see iterative improvements – more stable fungal circuits, standardized interfaces, and perhaps the first **real-world deployment** of fungal computers (for example, a fungi-based sensor network monitoring forest soil health or an intelligent fungal panel in a building adjusting ventilation based on detected air quality). Each success will illuminate new challenges, but also push the boundary of what an engineered living system can do.

6. Ethical and Philosophical Considerations

The advent of computing with living organisms – especially ones with potential intelligence like fungi – raises profound ethical and philosophical questions. Unlike traditional computers, **living substrates** have their own interests (survival, growth) and possibly some form of sentience or intrinsic value. Here we explore key considerations:

- **Intrinsic Value and Rights of Fungal Agents:** Should a fungal computer be treated merely as a piece of equipment, or as a form of life deserving ethical consideration? Traditionally, ethical frameworks in science cover animals (especially vertebrates) and sometimes plants, but fungi have rarely been considered – they were seen as insensate. However, emerging research

suggests fungi possess a kind of *proto-mind*. As biologist Nicholas Money argues, fungi “*operate as individuals, engage in decision-making, are capable of learning, and possess short-term memory*”³⁷. If we accept that fungi have a rudimentary consciousness or at least awareness⁷⁷,³⁸, it implies they exist on a continuum of cognitive life. This challenges the ethics of using them purely as tools. One might ask: is it **exploitation** to wire up a fungus and force it to compute? Of course, fungi don’t have a nervous system to feel pain in the way animals do, but they do respond to stress. For example, if a fungal computer is repeatedly shocked with electricity to stimulate it, is that causing harm? The notion of **consent** is not applicable in the human sense, but it urges scientists to consider minimizing unnecessary harm or stress to the organism. In practice, researchers are beginning to discuss treating the fungi in computing experiments with a level of care beyond that for inert materials – ensuring they have proper nourishment, avoiding lethal levels of stimulation, and providing rest periods. Some even speak of a “*duty of care*” for one’s experimental fungi, analogous to lab animal care, albeit at a simpler level. These ideas are not mainstream yet, but as fungal computing grows, we may see guidelines akin to bioethics for fungi.

- **Identity and Dignity:** A philosophical question arises: when we integrate fungi into our devices (say a fungal building or a fungal wearable), are we *instrumentalizing* a potentially intelligent being? There is a parallel to debates about AI and sentience – except here the “AI” is a living creature. Does a fungal network that is part of a building have any identity or dignity of its own? One interesting concept is **co-design with the organism**: architects in the FUNGAR project considered the fungi as a design partner, not just a material⁷⁸⁷⁹. They talk about a “living and thinking house made of fungi”⁷⁸ – implicitly acknowledging the fungus’s role as more than a passive component. Philosophically, this touches on **embodied cognition** and panpsychist ideas that mind-like properties might be ubiquitous in life. If one subscribes to those theories, then computing with fungi is essentially *harnessing a mind*. Some ethicists might argue this demands a new framework of respect, perhaps drawing from environmental ethics (valuing all living things) or from animal research guidelines (avoid suffering, use only when necessary, etc.). It’s a novel intersection: computing research and ethics of life.
- **Laboratory Ethics and Regulations:** Currently, fungi are not protected by animal research regulations. They are typically covered by biosafety rules (to prevent pathogens, etc.) but not by ethics boards regarding treatment. This could change if evidence of sentience mounts. We might imagine a future where causing undue harm to a complex fungal colony could be seen as ethically questionable, especially if the colony exhibits clear problem-solving or communication (signs of intelligence). On the flip side, some critics might view this as overreach – after all, we regularly harvest and eat mushrooms without ethical qualms. One could argue that using a fungus to compute is no more unethical than baking bread with yeast (which are fungi) or using bacteria to produce insulin. The counterpoint is that in computing we may be pushing these organisms to their limits in unnatural environments, effectively *creating a new class of lab organism*. There’s also the consideration of what happens when a fungal computer’s “job” is done – do we simply discard the living substrate (throwing away a life)? Or should there be some protocol (like reintroducing it to soil)? These are largely open questions awaiting societal and philosophical discourse.
- **Consent and Agency:** Obviously, fungi cannot give consent. But the notion of **agency** is worth pondering – the fungal network will be making decisions (computations) in the service of human-defined goals. Is there any sense in which the fungus has agency or is it entirely “programmed” by stimuli? Some philosophers like to explore whether by constraining a living being to perform computations, we are undermining its agency (like trapping an animal to perform tricks). While a fungus isn’t conscious in the animal sense, it does have a natural agenda

(seek nutrients, grow). We are co-opting that agenda when we use it for our computing tasks. Ensuring that the fungus can still carry out some of its natural behaviors (like growth and resource cycling) might be seen as respecting its agency in some minimal way. In practical terms, a well-fed fungus allowed to occasionally branch out might be a happier (and ironically, more *reliable*) computer than one kept starved in a tiny electrode grid purely for logic outputs.

- **Ethical Use-Cases:** We should also consider the ethics of *what we use fungal computers for*. For example, using fungal biocomputers in a **sustainable environmental monitoring** context (e.g. a network of fungal sensors that help protect a forest by detecting pollutants or fires) could be seen as an ethical positive – a symbiosis between technology and nature that benefits ecosystems. On the other hand, using fungi in a closed loop to optimize, say, high-frequency trading algorithms (a far-fetched scenario, but hypothetically) would be both impractical and ethically odd – repurposing life for a very artificial end. The most compelling vision, espoused by FUNGAR researchers, is *cohabitation*: intelligent buildings where humans and fungal networks live together symbiotically ⁸⁰ ¹². In such a scenario, ethical design might ensure the fungi get something out of the deal (nutrients, habitat) while humans get sensing and computing functions. It becomes a partnership rather than exploitation.
- **Public Perception and Biosecurity:** Another ethical aspect is how society perceives living computers. Some people may have an instinctual discomfort with the idea of living creatures inside their walls or devices (“my phone has a fungus brain!”). Transparent communication will be necessary to ensure public understanding and consent for technologies like fungal bio-buildings. There’s also a biosecurity angle: could engineered fungal networks pose environmental risks if they escape the lab? For instance, if we breed fungi for better electrical properties, we must ensure they’re not invasive or pathogenic if released. This is akin to GMO considerations. However, most fungi used (oyster, reishi) are common species and non-toxic, and likely the modifications are physical (embedding particles) rather than genetic at this stage. Still, careful thought is needed to not disrupt ecosystems – e.g. we wouldn’t want to unintentionally introduce a nanomaterial-laden fungus into wild populations.
- **Philosophical Insights:** On a more philosophical note, fungal computing challenges the mind-body dichotomy and anthropocentric views of intelligence. It invites questions like: *What is computation?* – If a colony of hyphae passing ions and nutrients around can compute, then computation is clearly an abstract process not limited to human-made devices. It also resonates with concepts of **embodied cognition and distributed mind**: the fungus processes information by *being* in the world (in soil, etc.), not by isolating itself from it. This could provide a model for “embodied AI” where intelligence isn’t just in a CPU but in the physical interaction of a system with its environment. Some have drawn parallels to the idea of the **“Wood Wide Web”** – the natural mycelial networks connecting forests – suggesting that fungal computers blur into the environment such that perhaps *the forest itself could one day be a computer*. If so, do we then attribute a form of consciousness or rights to an entire forest-network that we utilize for computing? These almost spiritual or eco-philosophical questions show how fungal computing links to **consciousness studies**: it expands the spectrum of systems we consider when pondering what consciousness is. If a mycelium can have a memory and adapt, is it a tiny bit conscious? What might that consciousness (or “fungal perspective”) be like? Some philosophers and scientists (e.g. those influenced by panpsychism or Whitehead’s process philosophy) might embrace fungal computing as evidence that *mind-like processes pervade nature*, just usually at slower, subtler scales. On a practical level, engaging with these questions ensures that as we develop bio-computers, we remain reflective about the implications.

In short, the ethics of fungal computing is nascent but multifaceted. It spans **welfare ethics** (treating the organism well), **environmental ethics** (doing it sustainably and safely), and **philosophical ethics** (respecting possibly conscious processes). As Andrew Adamatzky mused, we might have to learn to “**work in harmony with wetware**” rather than dominate it ⁸¹. This could mean designing computing systems that accommodate the fungi’s needs and rhythms – a mindset shift from treating them as just components. Such a paradigm is more *biocentric*, aligning with growing calls in technology for sustainability and respect for life. While we are far from according fungi legal rights (as we do animals in research), it is conceivable that guidelines will emerge (perhaps informed by fields like animal neuroscience ethics or plant ethics). For now, the focus is on awareness: researchers openly discuss these issues in their publications and outreach ⁸² ⁸³, priming both the scientific community and the public to approach fungal computing with a sense of responsibility and wonder. The philosophical insight we gain by literally *building a computer out of a living being* is that intelligence and computation might not be *ours* alone – they might be properties of life at large, inviting a more humble and interconnected view of technology.

7. Future Outlook and Emerging Directions

Fungal computing sits at the intersection of biology, computer science, and materials engineering – and its future will be shaped by advances in all three, as well as some far-out speculative ideas. Here are some expected and imagined developments beyond 2024:

Enhanced Fungal Strains and Synthetic Bio-Networks: One near-term direction is **bioengineering fungi for computing**. Just as researchers genetically tweak bacteria for logic gates, we may see efforts to modify fungi to improve their electrical properties or responsiveness. For example, one could engineer a fungus to express ion channels that make its spikes more pronounced or faster, thereby boosting its signal-to-noise ratio and speed. There’s precedent in other systems: scientists have made “*turbocharged*” neurons via gene editing; similarly, a fungus could be edited to overexpress certain membrane transporters to increase its electrical oscillation frequency. Another approach is to create **synthetic symbioses or biofilms** that combine organisms for better computing. Fungi naturally partner with algae (lichens) or plants (mycorrhizae) – perhaps a hybrid network of fungus + photosynthetic bacteria could compute in a way that’s self-powered (the bacteria provide energy from light, the fungus processes signals). There’s also interest in **mycelium-material composites** with embedded electronics (functional nanoparticles). If the FUNGAR doping experiments are extended, we might achieve a kind of **cyborg mycelium** – fungal tissue reinforced with conductive nanowires or graphene that enhance internal communication. This could produce a “*mycelium circuit board*” where the biological and artificial conductors intertwine at the microscopic level. By carefully patterning these additives, one could guide the formation of logic circuits within the living substrate (e.g. laying down a graphene grid that encourages the fungus to grow in specific pathways corresponding to electronic components). Such engineered networks blur the line between living and nonliving computing elements and might greatly improve performance while preserving self-healing and self-growing capabilities.

Quantum-Biological Links: In the far speculative realm, some researchers wonder if biological systems like fungi could interface with **quantum computing** or exploit quantum phenomena. It’s known that certain biological processes (photosynthesis, bird navigation, possibly even microtubule function) show quantum effects like coherence. Fungi have extensive **cytoskeletal networks (microtubules)** in their hyphae, which some theorists (in the spirit of Penrose and Hameroff) have proposed could support quantum computations or serve as biological qubits ⁸⁴. While this is unproven, one could imagine a future where engineered fungal cells have arrays of molecules that can exist in quantum superposition states, essentially acting as living qubits coupled to classical fungal electrical activity. This is highly speculative, but not entirely outlandish as the field of **quantum biology** grows. On a more practical

note, quantum sensors (like NV centers in diamonds) might be used to *monitor* fungal signals at the nanoscale, potentially reading single-ion channel activities. If we ever marry quantum computing hardware with a living network, we might get a **hybrid quantum-bio computer** where, say, a quantum processor handles certain calculations and passes results to a fungal network for analog processing or memory storage, leveraging the strengths of each (quantum speed vs. fungal memory).

Scale and Complexity – Toward Conscious Machines?: As we connect more living units together, questions arise about emergent complexity. A single mycelium has a sort of mind of its own – what happens if we network many mycelia? Do we get a “*meta-intelligence*” or just a larger substrate? Some futurists speculate about **planetary-scale computing** using living networks: for instance, augmenting the natural mycorrhizal networks in forests with sensors and communication nodes to create a distributed Earth-wide computer (literally Gaia as a computer). Even nearer term, a building with integrated fungal computing in all its walls and floors could act like a *brain for the building*, potentially exhibiting complex behaviors. Could such a system achieve a form of **consciousness**? This ties into embodied cognition theory – a building with sensory mycelium throughout might “feel” and adapt in a way that appears sentient (responding to occupants, weather, etc. in a learned manner). Whether that counts as consciousness is philosophical, but it certainly would challenge our current notions. Some draw parallels to the science-fiction concept of *living ships or habitats* – except here, it’s grounded in fungal computing networks rather than neural tissue per se.

Contemplative Intelligence and AI Synergies: There is growing interest in AI systems that incorporate slower, more **contemplative processes**. Modern AI (deep learning) is data-hungry and energy-intensive; incorporating living processors might offer a route to more efficient and perhaps more *cognitively interesting* AI. For example, one could imagine an AI that has a dual nature: a fast silicon-based module for rapid response and a slow fungal-based module for long-term deliberation. The fungal module, with its analog memory and gradual adaptation, could act as a kind of *reflective memory* or novel recurrent network that adds robustness. Already, reservoir computing is used in AI to handle temporal patterns – a fungal reservoir could imbue an AI with a kind of *long-term intuition* (albeit hard to interpret). There’s also an angle of **creative computing**: artists and composers have used slime mold and plant signals to generate music or visuals. Fungal rhythms might be tapped for creative AI, producing “organic” patterns that a purely digital algorithm might not. The term “**contemplative intelligence**” might also refer to AI that operates on human timescales or encourages human-like reflection. A fungal computer, by necessity, enforces patience and slower interaction, which could paradoxically be beneficial in certain AI contexts (for instance, encouraging an AI to gather more data over time instead of making a snap decision – something a fungus naturally does as it integrates conditions over hours). Such a system might interface better with human modes of thinking, even potentially facilitating a more *meditative human-computer interaction*. It’s speculative, but some envision technologies where working with a slow bio-computer becomes a form of mindfulness – you must slow down to its pace, leading to a more thoughtful use of tech (a counterpoint to today’s frenetic gadgets). This ties to philosophical movements in technology that advocate for “**slow tech**” or more humane tech experiences.

Beyond Fungi – Multi-Organism and Chimera Computers: Fungal computing might integrate with other unconventional platforms. We’ve discussed some (plants, slime, neurons, bacteria). A particularly intriguing direction is making **chimera systems**: for instance, coupling a fungal network with a silicon neural network or a cultured neuronal network to create a bio-hybrid computer that has multiple types of “intelligence.” Each part could handle what it’s best at – fungi for chemical sensing and memory, neurons for rapid computation, silicon for precise arithmetic – forming a symbiotic computing cluster. Imagine a robot whose control system includes a microprocessor for motor control, a small brain organoid for learning/adaptation, and a fungal mycelium for sensing the environment’s humidity and maintaining homeostasis. This robot might have capabilities and resilience unmatched by purely

engineered systems. We already see first steps: the slime mold smartwatch and fungal robots mentioned show we can plug these living pieces into functional devices. In the longer term, synthetic biology may produce entirely new network-like organisms specifically for computing. Think of a designed organism that grows like mycelium but with the fast electrical signals of neural tissue – a *synthetic neuromycelial network*. This could be the ultimate wetware: engineered life optimized for information processing, effectively a *living computer species*. While that sounds like science fiction, projects like **Physarum-based computing and “fungal machines”** (the title of a 2023 book ⁸⁵) are steps in that direction – exploring what a life form built for computing would look like.

Application Prospects: On the practical side, what applications might fungal (and related) biocomputers find? In the medium term, likely **environmental monitoring** and **sensor networks** will be the first real use. Fungal sensors could continuously monitor soil health, pollutants, structural strain in buildings, etc., with minimal maintenance. For example, a **mycelium-based structural health monitor** embedded in a bridge could detect cracks or stress (the fungus responds to new microcracks by changed electrical patterns ⁶⁶) and signal an alert. Because the fungus is self-repairing, it could survive harsh conditions better than electronic sensors that break. Another area is **medical or wearable biosensors**: since fungal mycelium can respond to chemicals, a small wearable with a living fungus might detect certain metabolites in sweat or the environment (some fungi react to specific airborne toxins by changing spikes frequency ¹³). Mycelium materials are already being used for wound dressings and skins; integrating computing into them could yield smart biomaterials that, say, sense infection and release a drug in response (the fungus could be engineered to produce an antibiotic when its electrical activity indicates the presence of pathogenic bacteria).

In the far future, **biocomputers** might be integrated into **ecosystems** for planetary-scale computing that assists in climate adaptation – for instance, guiding forest management by sensing and computing within the forest via fungal networks, essentially turning ecosystems into giant computers that help regulate themselves. This veers into the realm of **Gaian computing** and almost mysticism, but with climate change, such radical ideas of harnessing nature’s own networks for monitoring and response could gain traction.

Finally, from a scientific standpoint, pursuing fungal and bio computing will likely feed back into fundamental science: we will learn much more about how fungi communicate, make decisions, and maybe even how cognition could work in non-neural systems. It could influence **consciousness studies** by providing alternate models of sentience (e.g. a “fungal mind” vs an animal mind) and support theories that cognition does not require a brain or rapid spikes, only a network with feedback and adaptation. It also resonates with **embodied cognition**: intelligence as a process of the entire body interacting with environment, not just a central brain computing abstractly. Mycelial intelligence is exactly that – the whole network in the environment *is* the mind. Understanding it could lead to new computational paradigms that are more **context-aware and resilient** by design.

In summary, the future of fungal computing is richly interdisciplinary. We will see improvements in performance through bioengineering and hybridization with other tech (possibly even quantum elements). There will be increasingly practical deployments in environmental sensing and architecture, perhaps within this decade, given the momentum of projects like FUNGAR. Speculatively, the line between computing hardware and living systems will continue to blur – raising prospects of semi-living buildings, robots, and devices. This forces a rethinking of technology: rather than devices we manufacture and discard, they might be things we **grow, nurture, and even co-evolve with**. The journey of fungal computing from quirky lab demos to robust technology will undoubtedly bring surprises. As Karen Emslie wrote in 2023, breakthroughs are ranging “from bacteria-driven logic to disruptive technologies based on fungal mycelium – and even human brain cells” ⁸⁶. The convergence of these could herald a new paradigm where **“the computer is alive”** – not just metaphorically (as in AI)

but literally composed of living cells. With that comes incredible potential for sustainability and novel capabilities, but also the deep responsibility of creating and caring for computational systems that are, in a sense, *our living partners*. The next decade will test our ingenuity in engineering and perhaps our wisdom in handling this new form of technology.

Leading Researchers and Labs: It's worth noting a few key players driving these developments. Professor **Andrew Adamatzky** and the **Unconventional Computing Lab** at UWE Bristol remain pioneers, with prolific outputs on fungal, slime mold, and other biocomputers ⁸⁷ ¹³. Collaborators like **Han Wösten** (Utrecht University) bring mycology expertise to engineering fungi ⁹. The **FUNGAR consortium** united architects (Phil Ayres at CITA, Copenhagen ¹⁰), computer scientists, biophysicists, and industry (mycelium material companies like MOGU) – that cross-disciplinary team set a template for future projects. In the reservoir computing arena, groups like **Yuriy Kalinin's** lab (Boston) and European teams are blending live tissues with ML techniques. **Martyn Dade-Robertson** at Newcastle University has been exploring fungal architecture and synthetic biology integration (e.g. the **Living Architecture (LIAR)** project predated FUNGAR and looked at microbial fuel cells and computing in buildings). On the philosophical end, thinkers like **Nicholas Money** are expanding the view of cognition to fungi ⁸⁸. Startups such as **MycoSoft Inc** are emerging, showing entrepreneurial interest in fungal electronics. And not to forget, the work on neural organoids (Thomas Hartung at JHU and others) ties into the broader field of biocomputing that fungal research is now part of ⁵³. It's a small but rapidly growing community scattered across computing, biology, and design departments.

In Closing: Fungal computing began as a fringe curiosity – “mushroom logic gates? how peculiar!” – but from 2023 onward it is clearly maturing into a rigorous field with practical aspirations. The developments of the last two years have expanded fungal logic from a handful of gates to hundreds, integrated mycelium into architectural materials, improved interfaces, and even spun off nascent technologies. We can expect **new datasets** (for example, an update to Adamatzky's 470 logic functions library is anticipated as experiments continue – perhaps soon we'll see **thousands** of Boolean mappings catalogued under different conditions, shared as open data for the community). As leading labs refine techniques, we may get a “blueprint” of how to program a fungal network for a desired function, much like electronic circuit design. What was the stuff of science fiction – living computers, intelligent buildings – is increasingly supported by scientific evidence and prototypes ⁸⁰ ⁸. The future will likely not replace silicon chips with fungi in your laptop (fear not, your computer won't sprout mushrooms), but it will augment our technological landscape with **living, evolving computational media**. These media will excel in areas where conventional tech struggles: adaptability, integration with nature, and sustainable intelligence. Fungal computing is thus both a technological innovation and a paradigm shift – a reminder that intelligence and computation need not be confined to the machines we build, but can grow from the fabric of life itself.

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